Questions

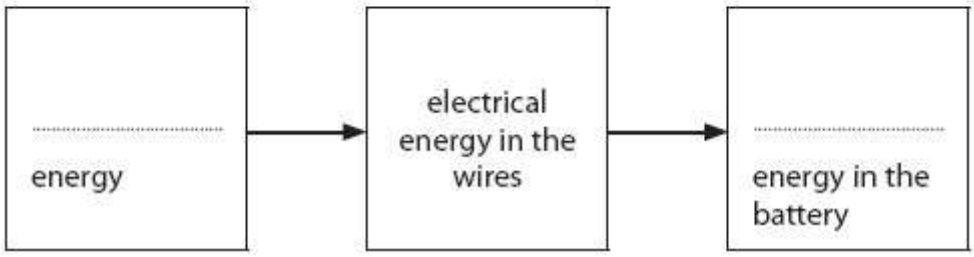
Q1.

A student uses a solar powered battery charger to charge some batteries.

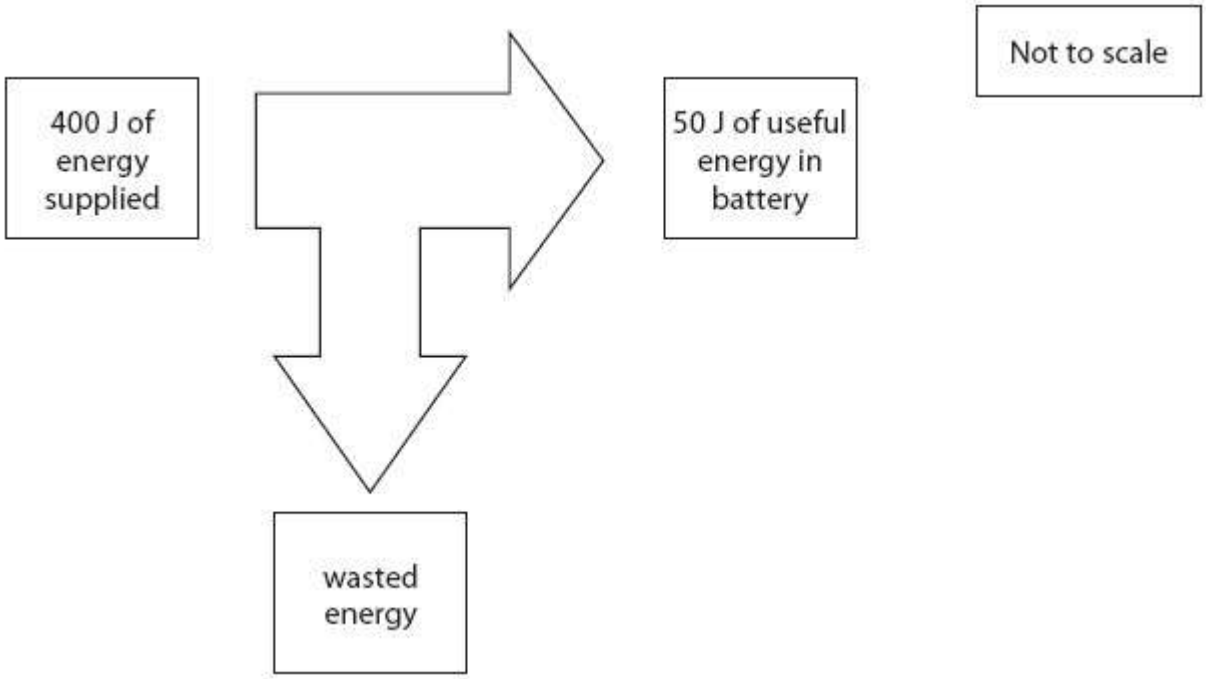
(a) The diagram is an energy transfer diagram for a battery being charged.
Use words from the box to complete the energy transfer diagram.

light	kinetic	sound
electrical	chemical	

Energy transfer diagram



(b) The diagram shows how much energy is usefully transferred by the battery charger.



(i) Calculate the amount of wasted energy.

(1)

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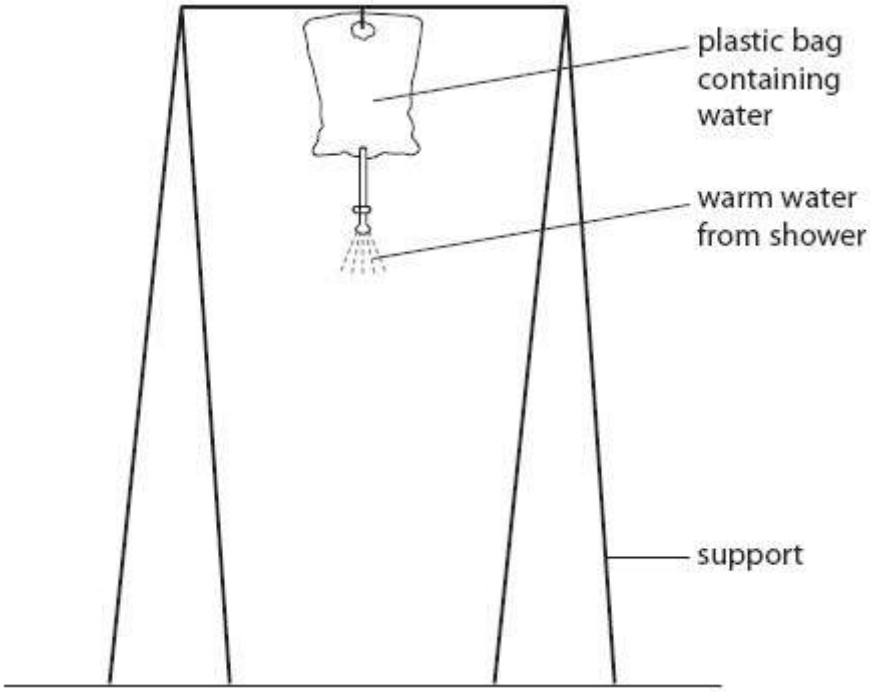
(ii) Calculate the efficiency of the battery charger.

(2)

.....

(c) The following arrangement is used as a solar powered shower.



The bag is left out in the sunlight during the day.

(i) Explain what colour the bag should be to heat the water to the highest temperature.

(2)

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(ii) On a sunny day the bag is filled with cold water.
Explain why the temperature of the water increases and then stays constant.

(3)

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(Total for Question = 10 marks)

Q2.

Figure 6 is a graph of the student's results.

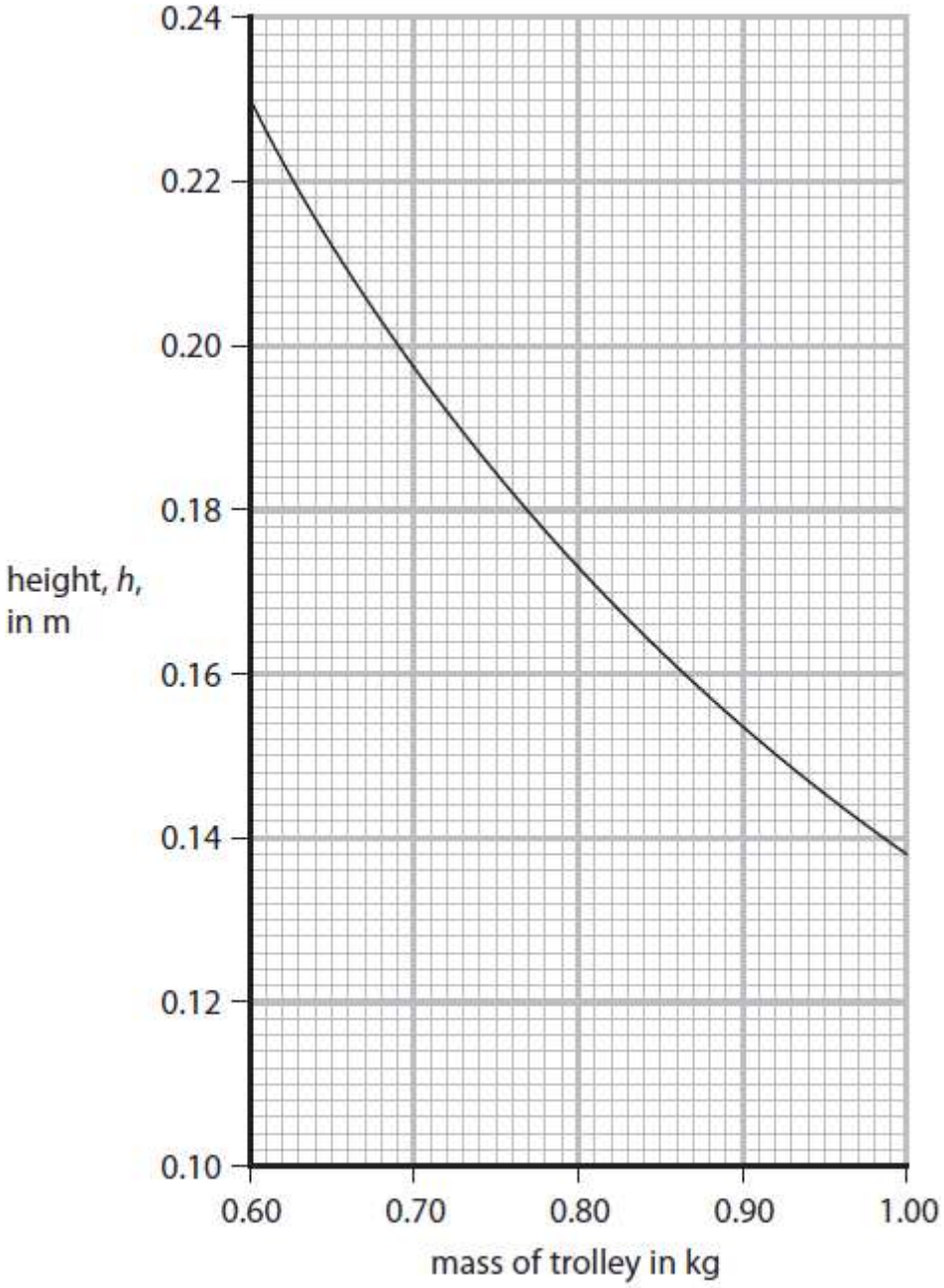

Figure 6

The student states that the energy transferred by the spring is the same each time it is used.

Use data from any two points on the graph in Figure 6 to support this statement.

(3)

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.....

(Total for question = 3 marks)

Q3.

Figure 7 shows a submarine being propelled forward underwater.

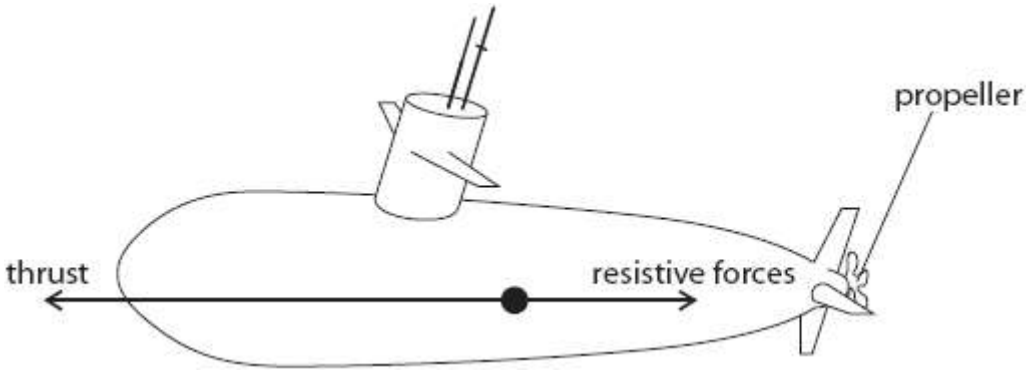

Figure 7

The thrust and the resistive forces are represented in magnitude and direction by the arrows in Figure 7.

Suggest **two** advantages of using nuclear-powered submarines compared to diesel-powered submarines.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q4.

Before a car brakes it has kinetic energy.
The kinetic energy decreases as it brakes.

State what happens to the kinetic energy during braking.

(1)

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.....

(Total for question = 1 mark)

Q5.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 2.

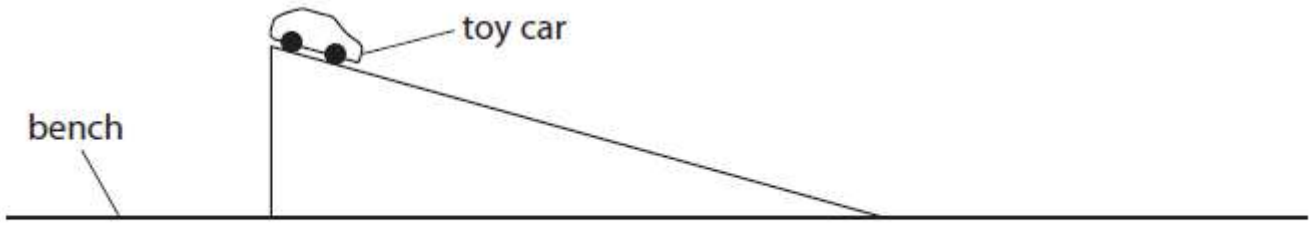

Figure 2

The student needs to develop the experiment to determine the loss in potential energy and the gain in kinetic energy as the toy car is rolling down the slope.

State the other measurements the student must make.

(2)

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.....

.....

.....

(Total for question = 2 marks)

Q6.

Figure 5 shows a way of projecting a small trolley up a sloping track.

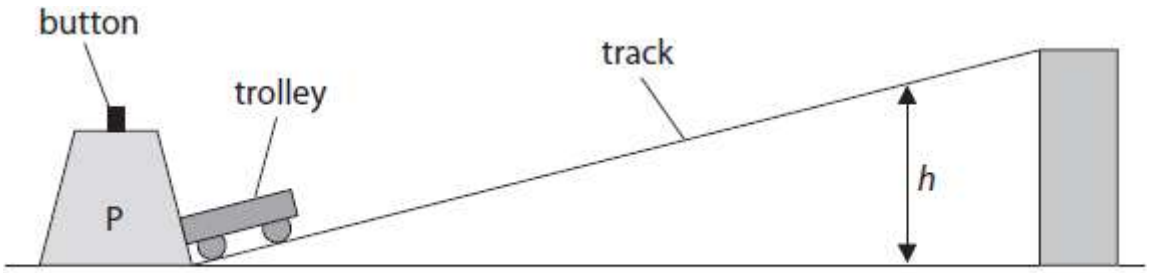

Figure 5

When the button is pressed, a spring is released in P that projects the trolley up the track.

The trolley travels up the track, stops and then rolls back down.

The spring in P always exerts the same force when projecting the trolley.

A student investigates how the mass of the trolley affects the maximum vertical height, *h*, reached by the trolley.

State the measurements the student should make to complete the investigation.

You should make use of the equipment shown in Figure 5 and any other equipment that is needed.

(4)

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(Total for question = 4 marks)

Q7.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 2.

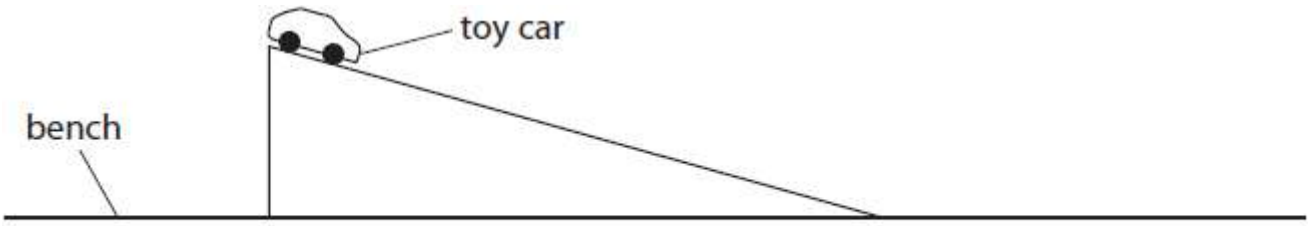

Figure 2

When the toy car rolls down the slope, some energy is transferred to the surroundings as thermal energy.

State how the student could calculate the amount of energy transferred to the surroundings.

(1)

.....

.....

(Total for question = 1 mark)

Q8.

Figure 2 shows a person on a skateboard at the top of a ramp.

At P, the person is not moving.

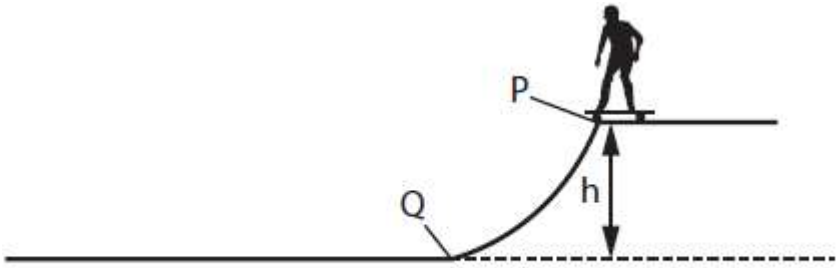

Figure 2

Figure 3 is a diagram that represents energy changes from P to Q.

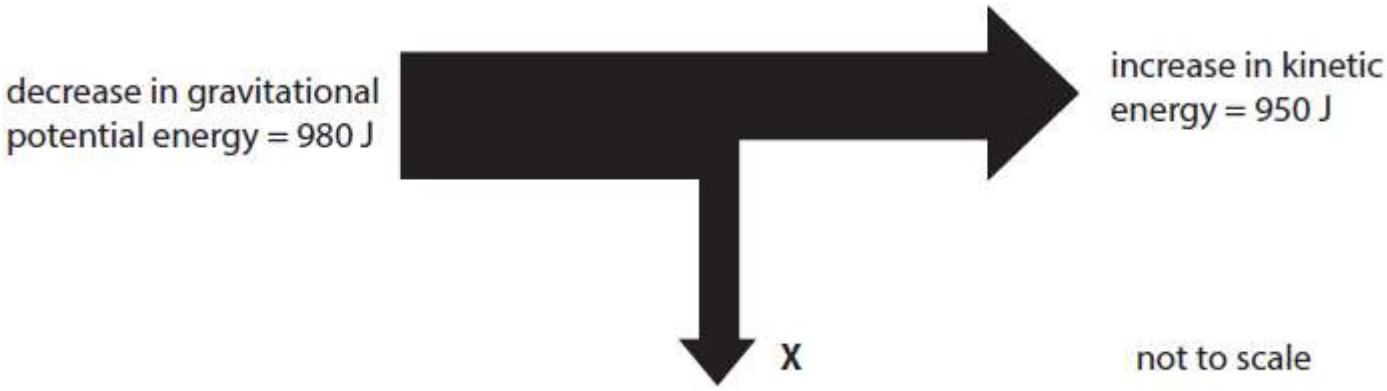

Figure 3

(i) State what is represented by **X**.

(1)

.....

.....

(ii) Calculate the value of **X**.

(1)

value of **X** = ... J

(iii) Calculate the efficiency of the system represented in Figure 3.

(2)

efficiency = ...

(Total for question = 4 marks)

Q9.

Explain why renewable sources provide an increasing fraction of the electricity supply for many countries.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(a)	**light** → electrical → **chemical** energy energy energy (1) (1)	These answers must be in the correct order	**(2)**
(b)(i)	350 (J)	400 – 50 (J)	**(1)**
(b)(ii)	Substitution 50 ÷ 400 (1) or $^{50 \times 100}/_{400}$ (%) Evaluation 13(%) (1)	12.5(%), 0.125, 0.13 or 1/8 Give full marks for correct answer, no working	**(2)**
(c)(i)	An explanation linking the following points: black (1) (because) (good) absorber (of thermal radiation) (1)	absorbs / takes in heat radiation **ignore** references to: attract good emitter light dark / darker	**(2)**
(c)(ii)	an explanation linking any **three** of the following points: • (bag / water) absorbs thermal energy / heat / radiation (1) • (bag / water) radiates / emits thermal energy / heat / radiation (1) • more heat radiated at higher temperature (1) • input and output are balanced (at steady temperature) (1)	idea of energy input e.g. "sun heats the bag up" idea of energy output idea of more heat lost (to surroundings)at higher temperature "absorbing heat at same rate as radiating heat" (3) **ignore** (sun) light / rays	**(3)**

Q2.

Question Number	Answer	Additional guidance	Mark
	reference to ΔPE = mg(Δ)h (1) relevant values from graph and one calculation to find energy (1) repeated with 2nd set of values (1)	can be seen in calculations e.g. 0.6 x 10 x 0.230 ≈ 1.4 (J) e.g. 1.0 x (10) x 0.138 ≈ 1.4 (J) must see calculations for mp2 and 3 1 mark for 2 calculations of mh with 'g' omitted (MP3)	**(3)**

Q3.

Question Number	Answer	Additional guidance	Mark
	Any **two** advantages such as: nuclear fuel • does not contribute to global warming • does not use up fossil fuels • occupies less space • does not involve a supply of air • gives less reason to surface/refuel • lasts longer	 accept reverse arguments does not produce carbon dioxide / greenhouse gases ignore cheaper, cleaner, greener, pollutes the sea, more efficient, unqualified safety, renewable	(2)

Q4.

Question Number	Answer	Additional Guidance	Mark
	one from causes heating of the surroundings (1) transferred to thermal energy of surroundings (1) increases the kinetic energy of molecules in the brake pads (1)	must include destination of final energy increases thermal energy of brake pads / wheels	**(1)**

Q5.

Question number	Answer	Additional guidance	Mark
	(vertical) height of slope (1) mass (of the toy car) (1)	 allow (in this context) weight if no other mark scored allow 1 mark for quoting either equation (Δ)GPE =mgh OR KE = ½ mv^2	**(2)**

Q6.

Question Number	Answer	Additional guidance	Mark
	An answer that includes: (measure) mass of the trolley (1) (measure) (vertical) height / h (1) repeat for a range of masses (1) plus **any one** from: method of identifying / measuring h (1) OR repeat firing with same mass (1)	 weigh the trolley NOT measure height of ramp e.g. use of reference mark accept "use ruler to measure height/h" for 2 marks NOT "use ruler to measure height of ramp"	**(4)**

Q7.

Question number	Answer	Additional guidance	Mark
	(original) GPE – KE (at bottom) (1)	allow (idea of) input – output allow wrong way round (eg output-input)	**(1)**

Q8.

Question number	Answer	Additional guidance	Mark
(i)	thermal (energy/store) (1)	energy dissipated energy wasted energy lost energy transferred to surroundings	**1** **AO2.1**
		energy transferred due to friction energy transferred due to air resistance allow heat (energy)	
		ignore sound / not useful energy	

Question number	Answer	Additional guidance	Mark
(ii)	30 (J) (1)	accept –30 (J)	**1** **AO2.1**

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) 950 (x100) 980		**2** **AO2.1**
	evaluation (1) 0.97 OR 97%	allow answers that round to 0.97 or 97% for 2 marks allow 1 mark for answers that round to 97 or 0.97% allow 1 mark for answers of 0.96 or 96% (truncated) award full marks for correct answer without working	

Q9.

Question Number	Answer	Additional guidance	Mark
	an explanation linking: increased use of renewables/decrease use of nonrenewables (1) reason (1)	 accept "them" as renewable accept reason why renewables are beneficial accept reason why non-renewable(s) are not beneficial	**(2)** AO 1 1